

CLINICAL MULTIPLE CHOICE EVALUATION

LECTURE 8 OF 20: PRESSURE ULCER PATHOPHYSIOLOGY, CATEGORISATION & PREVENTION

Candidate Instructions

- This exam paper contains **25 strict Multiple Choice Questions** matching the 2025 pressure ulcer curriculum.
- All questions explore the precise aetiology, numeric pressure benchmarks, classifications, and case studies in your slides.
- Select the single best answer choice (**A, B, C, or D**) for each question item.
- The full verification key matrix block is appended cleanly at the end of this booklet document.

QUESTION 1

According to the introductory slides, what basic physiological definition characterizes a pressure ulcer, also known as a bedsore or pressure injury?

- A) An allergic contact dermatitis breakout triggered exclusively by systemic heavy metal poisoning
- B) Localised damage to the skin and/or underlying tissue, typically occurring over bony prominences due to sustained pressure
- C) A superficial scrape caused by intense atmospheric cold that disrupts normal skin pH settings
- D) The rapid bacterial consumption of normal commensal flora within un-sterilized surgical lacerations

QUESTION 2

Under the core aetiology slide, how are 'Shear Forces' explicitly defined regarding tissue and skeletal movement?

- A) Friction that happens when the skin rubs against an external rough sheet surface, removing the stratum corneum
- B) Forces that occur when the skin moves in one direction while the underlying bone moves in another, damaging blood vessels
- C) Excessive moisture build-up from urine or sweat that macerates the outer dermis boundaries
- D) Sustained perpendicular compression that exceeds the metabolic baseline limits of cell mitochondria

QUESTION 3

What specific structural or physiological trait characterizes the effect of 'Friction' on skin tissues as listed on Slide 8?

- A) It compromises deep muscle fibers before any visible surface markings take place
- B) It causes superficial damage, such as abrasions or blisters, weakening the skin's protective barrier
- C) It generates sudden reactive oxygen species loops that provoke reperfusion oxidative injuries
- D) It causes an immediate drop in tissue interstitial pH fluid layers down to an acidic 2.5

QUESTION 4

According to Slide 7, sustained pressure exceeds standard 'capillary closing pressure' when it passes what explicit numeric benchmark?

- A) 12 mmHg
- B) 20 mmHg
- C) 32 mmHg
- D) 55 mmHg

QUESTION 5

During the ischemic cascade, hypoxia forces cells to shift their metabolic pathways. What is the explicit consequence of this metabolic shift as detailed in the text?

- A) A shift from anaerobic to aerobic pathways, causing high cholesterol synthesis structures
- B) A shift from aerobic to anaerobic processes, leading to the accumulation of metabolic waste products like lactic acid
- C) The immediate destruction of Langerhans cells, compromising the local skin immune response
- D) The rapid production of type III collagen fibers to thicken the surrounding basement membrane zone

QUESTION 6

Why does the slide outline that shear forces primarily affect deep tissues, such as muscles and subcutaneous layers, before visible skin changes occur?

- A) Because deep tissues have an un-blanchable redness trait that absorbs external light tracks
- B) Because they cause distortion and stretching of blood vessels at deeper anatomical planes
- C) Because the outer stratum corneum is completely immune to any mechanical tissue distortion
- D) Because deep muscle groups operate exclusively via anaerobic respiration at all shift times

QUESTION 7

What is the baseline biochemical range for normal skin pH, and what protective barrier layer does this range structurally create?

- A) pH 1.5 – 3.2; creates a sterile saline matrix
- B) pH 4.5 – 6.2; creates an acid mantle that acts as an effective neutralizing barrier
- C) pH 7.35 – 7.45; creates a trans-cellular fluid barrier
- D) pH 8.0 – 9.5; creates an alkaline lipid protective sheet

QUESTION 8

Why are older adults statistically more prone to moisture-related skin breakdown and friction injury according to Holroyd (2024)?

- A) Because their skin experiences an accelerated expansion of fat cell clusters inside the hypodermis
- B) Due to thinning of the epidermis, flattening of the dermis-epidermis junction, reduced elasticity, and skin atrophy
- C) Because their apocrine sweat glands generate massive levels of high-density glucose solutions
- D) Due to a sudden biological conversion of type I scar collagen strings back into type III loops

QUESTION 9

What specific paradoxical phenomenon occurs during 'Reperfusion Injury' when sustained tissue pressure is suddenly relieved?

- A) Lactic acid instantly shifts back into aerobic glycogen stores, halting cell necrotization
- B) The sudden influx of oxygen can generate reactive oxygen species (ROS), causing oxidative stress and additional damage
- C) Dermal capillaries immediately constrict to 0 mmHg, preventing any local oedema from forming
- D) The basement membrane zone separates completely from the spiny stratum spinosum tier

QUESTION 10

According to the NICE (2024) guideline criteria on Slide 14, who is classified as being at "high risk" of developing a pressure ulcer?

- A) Any individual over 50 years of age presenting with high-density skin tone characteristics
- B) People with multiple risk factors (e.g., significantly limited mobility, nutritional deficiency, history of PUs, or current PU)
- C) Patients presenting exclusively with localized transient blanchable erythema over the heel pad
- D) Individuals using high-density foam cushions for less than 30 minutes per calendar day

QUESTION 11

Which of the following contains the correct list of formal risk assessment scale tools named in your lecture syllabus?

- A) TIMERS, TILI, MUST, and the Waterlow Index card
- B) Braden Scale, Waterlow, Norton Risk assessment scale, and Braden Q Scale (for children)
- C) Datix Event Matrix, SCIE Map, and the ISTAP Skin Tear guidelines
- D) ABCDE approach, NEWS chart, and the BVM ventilation checklist

QUESTION 12

What specific clinical characteristics define a "Category 1" pressure ulcer?

- A) Partial-thickness loss of the dermis resulting in a shallow open ulcer with a pink wound bed and no slough
- B) Intact skin with non-blanchable redness of a localised area, usually over a bony prominence; area may feel painful, firm, soft, warmer or cooler
- C) Full-thickness tissue loss with visible subcutaneous fat, undermining, and tunneling paths
- D) A purple or maroon localized area of discoloration presenting as a blood-filled pocket blister

QUESTION 13

How does a "Category 2" pressure injury explicitly manifest visually according to Slide 16?

- A) Intact skin displaying transient blanchable erythema that shifts color when pressed
- B) Partial-thickness loss of the dermis resulting in a shallow open ulcer with a red/pink wound bed; no slough will be present; may also present as a serum-filled blister
- C) Full-thickness tissue loss where bone, tendon, and muscle are openly exposed to the floor elements
- D) An unstageable cavity fully obscured by thick black eschar and yellow tan slough structures

QUESTION 24

What are the precise anatomical boundaries and visibility limits laid out for a "Category 3" pressure ulcer?

- A) Partial-thickness epidermis wear where the underlying pink dermis layer is completely intact and unscratched
- B) Full-thickness tissue loss. Subcutaneous fat may be visible, but tendon, bone, or muscle will NOT be exposed; slough may be present
- C) Full-thickness tissue loss with exposed bone, tendon, and muscle, accompanied by a high probability of osteomyelitis
- D) Intact corneal layer thickening showing deep maroon discolored blood pockets

QUESTION 15

A "Category 4" pressure injury carries the explicit risk of which serious bone-level secondary complication due to its exposed structures?

- A) Pilosebaceous inflammation
- B) Osteomyelitis
- C) Acid mantle disintegration
- D) Reperfusion apoptosis cascade

QUESTION 16

How is an "Unstageable" pressure injury explicitly defined, and why is its true depth initially impossible to evaluate?

- A) It features intact skin with transient discolored patches that hide the underlying bone alignments
- B) Full-thickness tissue loss in which the base of the wound is obscured by either slough and/or eschar; true depth is unknown until enough is removed
- C) A partial-thickness scratch that is too small to be measured by standard hospital rulers
- D) A pediatric cerebral palsy lesion where limited mobility prevents positioning changes

QUESTION 17

What description perfectly matches a "Suspected Deep Tissue Injury" as presented on Slide 20?

- A) A shallow open ulcer with a bright red wound bed that presents with patchy yellow slough coatings
- B) Purple or maroon localised area of skin discolouration or blood-filled blister due to damage of underlying tissue from pressure/shear; may feel firm, mushy, boggy, warm or cool
- C) A well-repaired surgical incision line that demonstrates a sudden breakdown of type I collagen
- D) Superficial skin peeling caused by incontinence maceration across the stratum corneum

QUESTION 18

Under the pressure redistribution strategies, what is the specific operational rationale behind an 'Alternating Pressure Mattress'?

- A) It pumps high-volume hot air across the bedding to dry urine or sweat exudates
- B) They inflate and deflate sections to shift pressure points
- C) It conforms statically to contours without any internal cell mechanical movement
- D) It uses fluid gel layers to neutralize the skin's surface acid mantle

QUESTION 19

What explicit benefit and role is outlined on Slide 22 for a 'Low-Air-Loss Mattress'?

- A) It delivers direct electrical muscle stimulation to bedbound pediatric patients
- B) These maintain a dry environment and reduce friction
- C) It compresses capillary blood vessels to stop deep microvascular bleeding
- D) It completely replaces the clinical necessity for manual 2-hourly repositioning schedules

QUESTION 20

When executing standard clinical repositioning techniques, what specific anatomical areas are nurses directed to pay close attention to?

- A) The eyelids and the dorsal forearm quadrants
- B) Bony prominences (e.g., heels, sacrum, hips)
- C) The central abdomen and the large muscle beds
- D) The pre-tibial shin line and the cervical spine segments

QUESTION 21

Why is 'Adequate Protein' highlighted as an essential component within nutritional pressure ulcer prevention planning?

- A) It balances the body's trans-cellular fluid compartments to prevent edema from compression
- B) Protein is essential for tissue repair and regeneration; it helps maintain skin integrity and supports wound healing
- C) It acts as a direct antioxidant to prevent reactive oxygen species from formatting
- D) It elevates surface skin pH levels up to an alkaline neutral setting

QUESTION 22

Match the specific vitamins with their stated clinical care functions. Which statement is perfectly accurate based on Slide 24?

- A) Vitamin A acts as an antioxidant to block microvascular lactic acid formation
- B) Vitamin C promotes collagen synthesis and wound healing; Vitamin A supports epithelial tissue repair
- C) Vitamin E is responsible for initiating the clotting cascade over bony projections
- D) Vitamin D prevents shear force stretching by binding subcutaneous fat layers together

QUESTION 23

In Case Study 1, a 75-year-old woman admitted to a medical ward for heart failure develops a pressure ulcer on her sacrum. What did routine skin checks discover?

- A) A deep cavity with exposed sacral bone matching Category 4 severity criteria
- B) Non-blanchable erythema (Category I pressure ulcer)
- C) A shallow serum-filled blister displaying category 2 partial-thickness dermis wear
- D) A massive unstageable eschar patch covering her entire left hip quadrant

QUESTION 24

In Case Study 2, a 45-year-old man with paraplegia living independently at home undergoes self-monitoring. What pressure ulcer category does he discover on his left heel?

- A) Category I non-blanchable area
- B) A shallow ulcer (Category II)
- C) An unstageable slough crater
- D) Exposed heel tendon (Category IV)

QUESTION 25

Which advisory panel or institute is explicitly named alongside NICE in the guidelines section for offering evidence-based recommendations on pressure ulcers?

- A) The British Association of Critical Care Nurses (BACCN)
- B) The National and European Pressure Ulcer Advisory Panels (NPUAP / EPUAP)
- C) The International Skin Tear Advisory Panel (ISTAP)
- D) The Chronic Pain Policy Coalition of the UK

OFFICIAL EXAM ANSWER KEY MATRIX

Verify candidate choice selections strictly using this validation matrix block.

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	B	B	C	B	B	B	B	B	B

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
B	B	B	B	B	B	B	B	B	B

Q21	Q22	Q23	Q24	Q25
B	B	B	B	B